

1.10 Early Finishers: Sequences and Compound Interest — Solutions

Extension set, not scored: worked solutions only, no marks. Calculator allowed. Answers are exact or correct to three significant figures.

1. $u_3 + u_8 = 48, \quad u_{12} = 5u_2.$

(a) Write each term as $u_1 + (n-1)d$:

$$(u_1 + 2d) + (u_1 + 7d) = 48 \Rightarrow 2u_1 + 9d = 48$$

$$u_1 + 11d = 5(u_1 + d) \Rightarrow 6d = 4u_1 \Rightarrow u_1 = 1.5d$$

$$\text{Substitute: } 2(1.5d) + 9d = 48 \Rightarrow 12d = 48 \Rightarrow d = 4$$

$$u_1 = 1.5(4) = 6$$

Or rearrange both conditions to $2u_1 + 9d = 48$ and $4u_1 - 6d = 0$ and solve the system on the GDC. The sequence is 6, 10, 14, ...: $u_3 + u_8 = 14 + 34 = 48$ and $u_{12} = 50 = 5 \times 10$.

(b) $6 + 4(n-1) < 500 \Rightarrow 4(n-1) < 494 \Rightarrow n-1 < 123.5 \Rightarrow n < 124.5$

$u_{124} = 6 + 4(124-1) = 498$ and $u_{125} = 6 + 4(125-1) = 502$, so 124 terms are less than 500.

The terms below 500 are u_1 to u_{124} , so the count is the largest whole n , 124.

(c) The 1st, 2nd and 4th terms are 3, $3+d$, $3+3d$.

Geometric, so the ratios are equal: $\frac{3+d}{3} = \frac{3+3d}{3+d}$

$$(3+d)^2 = 3(3+3d) \Rightarrow 9+6d+d^2 = 9+9d \Rightarrow d^2-3d=0 \Rightarrow d(d-3)=0$$

$d \neq 0$, so $d = 3$.

The terms are 3, 6, 12, so $r = \frac{6}{3} = 2$.

$d = 0$ also solves the equation (the constant sequence 3, 3, 3), which is why the question excludes it.

2. Sofia: $PV = 5000$, 4.8% compounded monthly. Luca: $PV = 6000$, $r\%$ compounded quarterly, $FV = 9000$ after 10 years.

(a) $FV = PV \left(1 + \frac{r}{100k}\right)^{kn}$, with $k = 12$:

$$5000 \left(1 + \frac{4.8}{100(12)}\right)^{12n} > 10\,000$$

n (years)	Value
14	\$9777.63
15	\$10 257.42

15 years

Or TVM solver: $I\% = 4.8$, $PV = -5000$, $PMT = 0$, $FV = 10\,000$, $P/Y = C/Y = 12$ gives $N = 173.6$ months = 14.47 years; the value is first over \$10 000 at a whole year after 15 years. Show the adjacent year: 14 years is what shows 15 is the least.

(b) $6000 \left(1 + \frac{r}{100(4)}\right)^{4(10)} = 9000$

$$\left(1 + \frac{r}{400}\right)^{40} = 1.5$$

$$r = 400 (1.5^{1/40} - 1) = 4.0752 \dots \approx 4.08$$

TVM solver: $N = 40$, $PV = -6000$, $PMT = 0$, $FV = 9000$, $P/Y = C/Y = 4$, solve for $I\% = 4.0752 \dots$

(c) Sofia: $S_n = 5000 \left(1 + \frac{4.8}{100(12)}\right)^{12n}$ Luca: $L_n = 6000 \left(1 + \frac{4.0752 \dots}{100(4)}\right)^{4n}$

n (years)	Sofia	Luca
24	\$15 786.26	\$15 877.07
25	\$16 560.90	\$16 534.06

Year 25

Use a GDC table of both expressions, with the full-precision r stored from (b). Sofia starts with less but earns the higher effective rate ($1.004^{12} = 1.0491$, 4.91% a year, against Luca's $1.5^{1/10} = 1.0414$, 4.14% a year), so she catches up slowly. With $r = 4.08$ rounded, Luca's values are \$15 894.92 and \$16 553.42: the answer is still year 25, but carry full precision to the end.